

Birla Institute of Technology & Science, Pilani, Rajasthan
First Semester 2025-2026
Tutorial-2

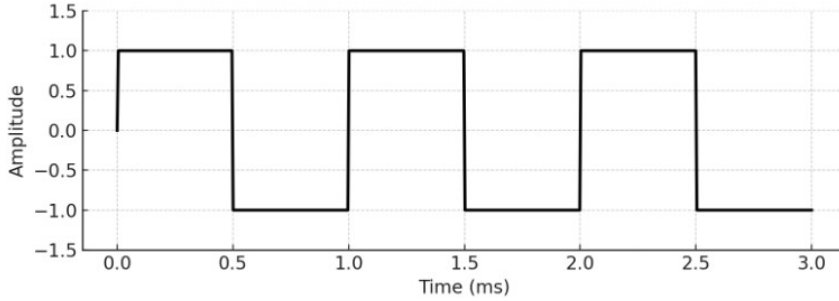
Course: ECE/EEE F311 Communication Systems
Instructor-in-Charge: Prof. Syed Mohammad Zafaruddin

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Objectives

- Fourier Series/Transform Applications, Distortion-less transmission, Auto-Correlation function.

Tutorial Problems



1. You are given a periodic square wave with period 1ms, as shown in figure. You also have signal processing blocks having built-in sinc function of any bandwidth. Draw a transmitter block to realize

$$x(t) = a \left[\sin(2\pi f_1 t) \right] \left[\sin(2\pi f_1 t) + \sin(2\pi f_2 t) \right] \left[\sin(2\pi f_1 t) + \sin(2\pi f_2 t) + \sin(2\pi f_3 t) \right]$$

where a is any constant with $f_1 = 1\text{KHz}$, $f_2 = 3\text{KHz}$, and $f_3 = 5\text{KHz}$. Also find the minimum channel bandwidth required for transmission.

2. *In communication systems, a channel is considered distortionless if it changes only the amplitude and timing of a signal without altering its shape. In other words, the output should be an exact replica of the input, except for a possible scaling in magnitude and a shift in time. This requires the channel to have constant gain and a phase response that varies linearly with frequency over the signal's bandwidth. Any variation in gain with frequency causes amplitude distortion, while any departure from a linear phase relationship leads to phase distortion.*

A signal $m(t)$ is transmitted through a channel of bandwidth B whose magnitude frequency response is given by

$$|H(f)| = \left[1 + k \cos(2\pi fT) \right]$$

with an additional nonlinear phase component described by

$$\theta(f) = -2\pi f t_d - k \sin(2\pi f T).$$

Derive the time-domain representation of the output signal $y(t)$ and examine the type and extent of distortion present when the parameter k is much smaller than unity ($k \ll 1$).

3. A signal

$$m(t) = 2 \cos\left(16\pi t - \frac{\pi}{4}\right) + \cos\left(40\pi t + \frac{\pi}{2}\right)$$

is transmitted through a distortionless channel. The received signal is observed to be

$$x(t) = 2 \cos\left(16\pi t + \frac{3\pi}{4}\right) + \cos(40\pi t + \phi).$$

Determine the minimum channel delay t_d and the phase ϕ of the second component.

4. Find the auto-correlation function $R_x(\tau)$ and energy/power spectral density of the following signals

- $x(t) = e^{-at}u(t)$
- $x(t) = u(t)$

Q4: if time permits.

Solution

1. The **Fourier series** for the 50% duty-cycle square wave (no derivation shown) is

$$m(t) = \frac{4A}{\pi} \left[\sin(2\pi(1 \text{ kHz})t) + \frac{1}{3} \sin(2\pi(3 \text{ kHz})t) + \frac{1}{5} \sin(2\pi(5 \text{ kHz})t) + \dots \right].$$

To construct $x(t)$:

- (a) Pass $m(t)$ through a transmitter filter $2000 \text{ sinc}(2000\pi t)$ to get the 1 kHz component, yielding $x_1(t)$.
- (b) Pass $m(t)$ through a transmitter filter $6000 \text{ sinc}(6000\pi t)$ to the signal upto 3 kHz component, yielding $x_2(t)$.
- (c) Pass $m(t)$ through a transmitter filter $10000 \text{ sinc}(10000\pi t)$ to get the signal upto 5 kHz component, yielding $x_3(t)$.

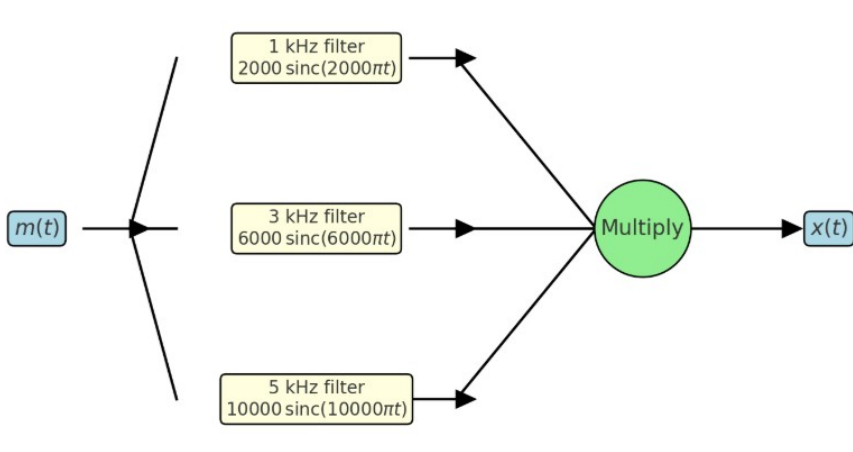
Operate all three branches in parallel, then **multiply** the outputs to obtain:

$$x(t) = x_1(t) \cdot x_2(t) \cdot x_3(t).$$

Since the multiplication produces sum-frequency terms, the **highest frequency** is:

$$f_{\max} = 1 + 3 + 5 = 9 \text{ kHz}.$$

Thus, the **minimum channel bandwidth** required is **9 kHz**.



- 2.

$$H(f) = [1 + k \cos(2\pi fT)] e^{-j2\pi f t_d} e^{-jk \sin(2\pi fT)},$$

$$\cos(2\pi fT) = \frac{1}{2} (e^{j2\pi fT} + e^{-j2\pi fT}), \quad e^{-jk \sin(2\pi fT)} = \sum_{n=-\infty}^{\infty} J_n(k) e^{-j2\pi f n T}.$$

where J_n is the Bessel function.

$$H(f) = e^{-j2\pi f t_d} \left(\sum_{n=-\infty}^{\infty} J_n(k) e^{-j2\pi f n T} \right) \left(1 + \frac{k}{2} e^{j2\pi fT} + \frac{k}{2} e^{-j2\pi fT} \right).$$

$$H(f) = e^{-j2\pi f t_d} \left[\underbrace{\sum_{n=-\infty}^{\infty} J_n(k) e^{-j2\pi f n T}}_{(A)} + \frac{k}{2} e^{j2\pi f T} \underbrace{\sum_{n=-\infty}^{\infty} J_n(k) e^{-j2\pi f n T}}_{(B)} + \frac{k}{2} e^{-j2\pi f T} \underbrace{\sum_{n=-\infty}^{\infty} J_n(k) e^{-j2\pi f n T}}_{(C)} \right]. \quad (1)$$

(A) No shift: Let $m = n$

$$\sum_n J_n(k) e^{-j2\pi f n T} = \sum_m J_m(k) e^{-j2\pi f m T}.$$

(B) $+T$ factor:

$$e^{j2\pi f T} \sum_n J_n(k) e^{-j2\pi f n T} = \sum_n J_n(k) e^{-j2\pi f (n-1) T}, \quad (2)$$

$$m = n - 1 \quad \Rightarrow \quad n = m + 1, \quad (3)$$

$$= \sum_m J_{m+1}(k) e^{-j2\pi f m T}. \quad (4)$$

(C) $-T$ factor:

$$e^{-j2\pi f T} \sum_n J_n(k) e^{-j2\pi f n T} = \sum_n J_n(k) e^{-j2\pi f (n+1) T}, \quad (5)$$

$$m = n + 1 \quad \Rightarrow \quad n = m - 1, \quad (6)$$

$$= \sum_m J_{m-1}(k) e^{-j2\pi f m T}. \quad (7)$$

Combine all terms:

$$H(f) = \sum_{m=-\infty}^{\infty} \left[J_m(k) + \frac{k}{2} (J_{m+1}(k) + J_{m-1}(k)) \right] e^{-j2\pi f (t_d + m T)}. \quad (8)$$

Using $e^{-j2\pi f a} \longleftrightarrow \delta(t - a)$,

$$h(t) = \sum_{m=-\infty}^{\infty} \left[J_m(k) + \frac{k}{2} (J_{m+1}(k) + J_{m-1}(k)) \right] \delta(t - t_d - m T)$$

Thus,

$$y(t) = m(t) * h(t) = \sum_{m=-\infty}^{\infty} \left[J_m(k) + \frac{k}{2} (J_{m+1}(k) + J_{m-1}(k)) \right] m(t - t_d - m T).$$

For $k \ll 1$, use the small-argument Bessel approximations:

$$J_0(k) \approx 1, \quad J_{\pm 1}(k) \approx \pm \frac{k}{2}, \quad J_{|m| \geq 2}(k) \approx 0.$$

$$m = 0: \quad J_0 + \frac{k}{2} (J_1 + J_{-1}) \approx 1 + \frac{k}{2} \left(\frac{k}{2} - \frac{k}{2} \right) = 1,$$

$$m = 1: \quad J_1 + \frac{k}{2} (J_2 + J_0) \approx \frac{k}{2} + \frac{k}{2} (0 + 1) = k,$$

$$m = -1: \quad J_{-1} + \frac{k}{2} (J_0 + J_{-2}) \approx -\frac{k}{2} + \frac{k}{2} (1 + 0) = 0.$$

$$y(t) \approx m(t - t_d) + k m(t - t_d - T).$$

For small k the channel introduces a post-echo of strength k at delay T .

3. Input signals:

Start time of first sinusoidal $16\pi t - \pi/4 = 0$ gives $t = 1/64$. Start time of the second sinusoidal $-1/80$.

Output signals:

Start time of first sinusoidal $16\pi t + 3\pi/4 = 2\pi$ gives $t = 5/64$. Note that using 0 instead 2π will give t negative.

Delay introduced by the channel $t_d = 5/64 - 1/64 = 1/16$.

The delay of the second component should be the same:

$$40\pi \left(-\frac{1}{80} + \frac{1}{16} \right) + \phi = 0$$

gives $\phi = -2\pi$.

4. • $x(t) = e^{-\alpha t} u(t)$.

$$X(f) = \frac{1}{\alpha + j2\pi f}$$

Energy spectral density

$$\mathcal{G}_X(f) = |X(f)|^2 = \frac{1}{\alpha^2 + 4\pi^2 f^2}.$$

Thus

$$R_X(\tau) = \mathcal{F}^{-1}[\mathcal{G}_X(f)] = \frac{1}{2\alpha} e^{-\alpha|\tau|}.$$

•

$$\begin{aligned} R_X(\tau) &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} u(t) u(t - \tau) dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{\tau}^{\frac{T}{2}} dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \left(\frac{T}{2} - \tau \right) = \frac{1}{2} \end{aligned}$$

Thus,

$$S_X(f) = \mathcal{F}[R_X(\tau)] = \frac{1}{2} \delta(f).$$